Resolution Geometry

Physical Curvature:
The Canonical Connection as a Law of Motion, Duration and Load

Final edition (supersedes the first edition and its lock supplement)

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Abstract

For a regular partially observed system with a declared positive cost P, Resolution Geometry (RG) selects the unique minimum-P^{-1} horizontal lift $B = PJ^\top(JPJ^\top)^{-1}$. We show that this canonical connection is a *law of motion* whenever the response hypothesis H-resp holds, and H-resp holds exactly in two classes of systems:

- overdamped (Stokes) systems, where P is the mobility;
- free bodies with zero conserved momentum in the unobserved directions, where P is the inverse inertia.

The connection then has three layers of physical content.

1. *Geometry.* Holonomy of closed observed loops is oriented area times curvature, to leading order. The paper treats an exact apparatus (residual 9.9995 μm at $L = 100$ mm), the three-sphere microswimmer (curvature $(a/3)[L_1^{-2} + L_2^{-2} - (L_1 + L_2)^{-2}]$), and a falling-cat body (-0.2748 rad per joint loop). Reciprocal strokes cancel exactly, which is Purcell's theorem. Zero total signed area does *not* imply zero holonomy.
2. *Duration and energy.* With an absolute mobility and a power anchor W, the minimum duration of a stroke is $\ell/\sqrt{W}$ and its minimum energy at duration T is ℓ^2/T, where ℓ is the dissipation length of the lifted path. Unloaded displacement is rate-independent. The connection alone supplies no absolute clock.
3. *Load.* The response is affine, $v = B\dot{q} + (I - BJ)Pf_{\text{ext}}$. For the swimmer the load term depends only on the total applied force, which yields a linear load law and a predicted stall force. Nonzero conserved momentum adds a dynamical drift to the geometric phase.

Two evidence layers test these laws within declared models.

- *Original battery* (11 checks, of which 10 are physical). Geometry fitted to *passive* Brownian fluctuations predicts *active* swimming, within 0.15–6.4% over ten runs.
- *Frozen holdout.* Separate processes calibrate, predict and score. On 24 held-out strokes the maximum errors are 0.47% for displacement and 0.066% for duration, and the stall residual is 2.3×10^{-6}. All 18 checks pass, and 10 of 10 canaries and controls are caught.

An independent ten-check audit, sharing no code with either battery, reproduces every key law. This is RG doing physics with its own geometry, conditionally, on synthetic models. It is not spacetime curvature or laboratory validation, and it claims no priority over the gauge theory of deformable bodies.

Contents

1 The standard and the dependency discipline **2**

2 The connection, its curvature, and the local falsifier **3**

3 An exact apparatus **3**

4 When the connection is a law of motion **4**

5 Duration and energy **4**

6 Load and momentum **5**

7 Three physical systems **5**
7.1 The three-sphere microswimmer 5
7.2 A falling-cat body 6
7.3 Fluctuations predict locomotion 6

8 Frozen holdout **6**

9 What is RG's, and what is not **7**

10 Experimental programme **8**

11 Status **8**

A Errata to the first edition **9**

B Reproducibility **9**

1 The standard and the dependency discipline

To have "business in physics", a framework must turn an object it defines into falsifiable predictions about motion, from inputs calibrated without the predicted effects. The layers of this paper are kept separate:

1. *Mathematics,* conditional on full row rank of J, $P \succ 0$, smoothness, and the stated response hypotheses.

2. *Physical inputs:*

   - the observation map F;
   - the mobility or kinetic metric;
   - the force or momentum constraints;
   - for duration, an absolute power or clock anchor.

   Each of these is locally calibrated.

3. *Numerical evidence:* synthetic tests of declared models (Oseen hydrodynamics, point masses) and their implementations. No laboratory data are used.

4. *Non-claims:* no spacetime curvature, gravitational law, universal time, quantization, model-independent hydrodynamics, or priority over established geometric mechanics.

The object is RG's canonical latent connection (print ladder L80). Its non-selection without P is ladder L79.

2 The connection, its curvature, and the local falsifier

Theorem 2.1 (Canonical lift)**.** *For J of full row rank and $P \succ 0$, the unique minimizer of $\frac{1}{2}w^\top P^{-1}w$ subject to $Jw = v$ is $w = Bv$, with*

$$B = PJ^\top(JPJ^\top)^{-1} \quad \textit{and} \quad JB = I.$$

Let $V = I - BJ$ be the P^{-1}-orthogonal projection onto $\ker J$.

Proof. By Lagrange multipliers and strict convexity. Numerically, the formula agrees with constrained optimization on 200 random instances (PC01). □

Definition 2.2 (RG transport curvature)**.** With $H_i = Be_i$, define $\Omega_{ij} = V[H_i, H_j]$, the Ehresmann (O'Neill) curvature [2, 3], valued in $\ker J$ ($J\Omega = 0$).

Theorem 2.3 (Leading-order holonomy)**.** *For an observed rectangle with both sides $O(\varepsilon)$, traversed $1 \to 2 \to -1 \to -2$, every observation returns exactly, and the unobserved endpoint moves by $ab\,\Omega_{12} + O(\varepsilon^3)$. This is a local expansion, not an exact finite-loop area formula. Reversal gives the* inverse holonomy map*. A sign-reversed displacement is guaranteed at leading order, or exactly in Abelian, symmetry-equivariant realizations; it is not a general finite-loop vector identity.*

No-Go 2.4 (The observation map alone does not fix the curvature; PC03). For the same F, a commuting lift exists, and it gives zero holonomy. The curvature is locked by $(F, P,$ rule), not by observational equivalence (L79).

Proposition 2.5 (Covariance; PC04)**.** *Under a nonlinear latent reparametrization, transforming P as a tensor preserves the physical endpoint (difference 4×10^{-15}). Resetting P to the identity in the new chart changes it (1.2×10^{-3}).*

Proposition 2.6 (Local falsifier of H-resp)**.** *H-resp is testable without closing any loop. Measure the response v to a commanded $\dot{q}$, then check both $Jv = \dot{q}$ and $Vv = 0$ within a declared error budget. Agreement of shape rates alone is insufficient: adding a common translation $0.01(1, 1, 1)$ to a swimmer response leaves the shape rates unchanged, but produces the vertical residual $0.01\sqrt{3} = 0.0173205$.*

3 An exact apparatus

Take configuration (X, Y, Z) and feedback channels $U = X$ and $V = Y + XZ/L$, with an isotropic cost. Then

$$\Omega_{uv} = \frac{(0, -x, 1)}{(1 + x^2)^2},$$

and for an observed rectangle of sides A, B,

$$\Delta Z = \frac{AB}{\sqrt{L^2 + A^2}} = \frac{AB}{L} - \frac{A^3 B}{2L^3} + O\left(\frac{A^5 B}{L^5}\right).$$

For $L = 100\,\text{mm}$ and $A = B = 1\,\text{mm}$, the prediction is $\Delta Z = 9.9995000375\,\mu\text{m}$. The correction is cubic in A at fixed B, but quartic when $A = B = O(\varepsilon)$, so this example cancels the generic cubic remainder. Integration agrees to machine precision, and the linear control gives zero (PC02, PC10).

No-Go 3.1 (Zero signed area is not flatness). Take two 0.15×0.15 lobes of opposite orientation, joined by an out-and-back connector of length 0.8. The total signed area is 0, and both observations return to within 1.2×10^{-16}, yet $\Delta Z = 0.0126429058$. Opposite lobes sample different curvature. This was reproduced independently (A01).

Protocol. The experiment proceeds in four steps:

1. calibrate J and P by single-leg responses;
2. test H-resp locally;
3. register ΔZ;
4. run the loop.

4 When the connection is a law of motion

Hypothesis 4.1 (H-resp)**.** The system responds to a requested observed velocity with the minimum-P^{-1} lift.

Theorem 4.2 (Physical realization)**.** *H-resp holds exactly in two classes of systems:*

(a) Overdamped systems *with* $\Gamma\dot{\xi} = f$ *and no external force on* $\ker J$*. Then* $f \in \operatorname{im} J^\top$*, and* $P = \Gamma^{-1}$ *is the mobility. The motion is the minimum-dissipation lift [5, 6].*

(b) Free bodies *whose momentum conjugate to the unobserved symmetry directions is conserved and zero. Then* $M\dot{\xi} \in \operatorname{im} J^\top$*, and* $P = M^{-1}$*. The motion is the mechanical connection [16, 15].*

Proof. In both cases $P^{-1}\dot{\xi} = J^\top\lambda$ and $J\dot{\xi} = \dot{q}$, which are the Lagrange conditions of the canonical lift. □

Corollary 4.3 (Estimation connection equals physical connection)**.** *The Einstein relation gives diffusion covariance* $2k_BT\Delta t$ *times the mobility [7]. Since* B *is invariant under* $P \to cP$*, the covariance-weighted (noiseless Kalman) lift is the physical lift [4]. Temperature cancels from the* connection*. It does not cancel from absolute timing (Section 5).*

5 Duration and energy

In this section P is an *absolute* mobility, so P^{-1} is a resistance tensor. The dynamics is overdamped, instantaneous, unloaded and memoryless. Fix an observed path and its horizontal lift $x(u)$, $0 \le u \le 1$; scheduling does not change the lift.

Theorem 5.1 (Dissipation, duration and energy)**.** *Let* $G = (JPJ^\top)^{-1}$*,* $h(u) = q'^\top G q'$*, and* $\ell = \int_0^1 \sqrt{h}\,du$*. The dissipated power is* $\mathcal{P} = \dot{q}^\top G\dot{q}$*. Then:*

- *under a power cap* $\mathcal{P} \le W$*, the minimum duration is* $T_{\min} = \ell/\sqrt{W}$*;*
- *at a prescribed duration* T*, the minimum dissipated energy is* $E_{\min} = \ell^2/T$*.*

Constant dissipation-metric speed attains both.

Proof. First, $v^\top P^{-1} v = \dot{q}^\top G\dot{q}$. With $r = du/dt$ we have $\mathcal{P} = hr^2$, so the cap gives $dt/du \ge \sqrt{h/W}$. Since $\ell = \int \sqrt{\mathcal{P}}\,dt$, Cauchy–Schwarz gives $\ell^2 \le TE$. □

Units: with length coordinates, $[P] = \mathrm{m}/(\mathrm{N\,s})$, $[\ell] = \sqrt{\mathrm{J\,s}}$ and $[W] = \mathrm{J/s}$, so $[\ell/\sqrt{W}] = \mathrm{s}$ and $[\ell^2/T] = \mathrm{J}$. The checks agree:

- a brute-force optimal schedule reproduces ℓ^2/T to 3×10^{-6} (A03);
- a power-controlled force ODE reproduces the metric-integral time to 3.5×10^{-11} (EX18);
- the uniform schedule costs 1.069 times the optimum (EX06).

Proposition 5.2 (No absolute clock from the connection; EX04, A04)**.** *Replacing P by cP leaves B and every lifted path unchanged, but sends ℓ to $\ell/\sqrt{c}$ and $T_{\min}$ at fixed W to $T_{\min}/\sqrt{c}$. Absolute time therefore needs an independently anchored mobility prefactor (for example $2k_BT\Delta t$) and a power or clock anchor. RG's duration layer (the ladder's clock rungs) supplies the anchoring discipline. The connection does not supply it.*

Remark 5.3 (The dissipation metric need not descend; EX17, A10)*.* G lives along the declared latent lift. For $F = x$ and $P = \mathrm{diag}(1+z^2, 1)$, it takes the values 1 and $1/5$ at the same observation. Quotient-level duration therefore requires basicness or a declared state rule, and ℓ is not a geodesic distance between observed endpoints.

Corollary 5.4 (Rate independence; EX07, A07)**.** *Unloaded displacement per cycle is invariant under monotone reschedulings. It agrees under $1\times$, $3\times$ and warped schedules to 10^{-12}, so the mean velocity is $\Delta X/T$. Stretching a schedule by a factor c scales power by c^{-2} and energy by c^{-1}.*

6 Load and momentum

Proposition 6.1 (Affine overdamped response)**.** *If $P^{-1}v = J^\top\lambda + f_{\mathrm{ext}}$ and $Jv = \dot{q}$, then $v = B\dot{q} + (I - BJ)Pf_{\mathrm{ext}}$. This agrees with a direct KKT solve to 10^{-15} (A05).*

Corollary 6.2 (Load law and stall force)**.** *For the translation-invariant three-sphere model,* $\ker J = \mathrm{span}\{\mathbf{1}\}$*, and the vertical part of $(I - BJ)Pf_{\mathrm{ext}}$ is $\mathbf{1}\, f/(\mathbf{1}^\top P^{-1}\mathbf{1})$, which depends only on the* total *applied force f. Hence*

$$\Delta X(f) = \Delta X(0) + fK, \qquad K = \int_0^T \frac{dt}{\mathbf{1}^\top P(q(t))^{-1}\mathbf{1}}, \qquad f_{\mathrm{stall}} = -\frac{\Delta X(0)}{K}.$$

Unlike unloaded holonomy, the load term grows with duration. In the independent audit, a direct force-balance swimmer loaded on a single bead reproduces the slope K to 10^{-9} and stalls at the predicted force with residual 3×10^{-13}. Tripling the duration triples the load increment (A06, A07).

Proposition 6.3 (Nonzero conserved momentum; EX09, A08)**.** *For the planar chain with conserved angular momentum L,*

$$\dot{\theta} = -\frac{M_{\theta s}\dot{s}}{M_{\theta\theta}} + \frac{L}{M_{\theta\theta}}.$$

The first term is the zero-momentum connection and the second is a dynamical drift. The formula reproduces Cartesian momentum at 30 states, with error at most 3.9×10^{-11} (battery) and 2.6×10^{-10} (audit). A loop with $L \neq 0$ has both geometric and time-dependent parts.

7 Three physical systems

7.1 The three-sphere microswimmer

Three beads of radius a on a line, with Oseen interactions, form a force-free swimmer, so H-resp(a) holds. To first order in a, the RG curvature is purely translational:

$$\Omega = \frac{a}{3}\left[\frac{1}{L_1^2} + \frac{1}{L_2^2} - \frac{1}{(L_1+L_2)^2}\right].$$

This bracket has the functional form of the known leading-order swimming law [11, 12]. The coefficient is derived here, and validated against an independent force-free simulation (PC05):

a	d	simulated	RG prediction	ratio
0.05	0.1	9.584×10^{-4}	9.163×10^{-4}	1.046
0.025	0.05	1.169×10^{-4}	1.145×10^{-4}	1.020
0.0125	0.025	1.445×10^{-5}	1.432×10^{-5}	1.010

Every bead is displaced equally, so the holonomy is a group element (PC09).

Corollary 7.1 (Reciprocity, the scallop theorem; PC06)**.** *A reciprocal stroke retraces its path, so its transport is followed by the inverse transport, and it cancels exactly even where the curvature is nonzero. In-phase strokes and the two-sphere swimmer give* $|\Delta| < 10^{-15}$ *[8]. This is* cancellation on reciprocal paths*, not flatness (compare the zero-area No-Go). Without hydrodynamic interactions, the connection* is *flat and nothing swims at all.*

7.2 A falling-cat body

A free planar three-link chain carries zero angular momentum; it has four unit masses and links of 0.3 m. The RG lift with $P = M^{-1}$ equals the angular-momentum law to 2×10^{-10}. A 0.8×0.8 rad joint loop based at $(1.5, 1.5)$ reorients the body by -0.2748 rad. Rotations commute, so the same square traversed backwards gives $+0.2748$ rad exactly. The rotation per area converges to the curvature, and a lift with P the identity predicts no rotation and is falsified (PC07) [14, 15].

7.3 Fluctuations predict locomotion

The connection is fitted to passive Brownian increments by regression at 81 shapes, with no driving and no loop. It predicts the active per-cycle displacement within 0.95–1.52% on the five original seeds, and within 0.15–6.37% on five further seeds. All ten runs meet the registered 15% gate. The accuracy is run-specific, not a general guarantee. Calibrating without hydrodynamic interactions predicts no swimming, and the canary catches it (PC08).

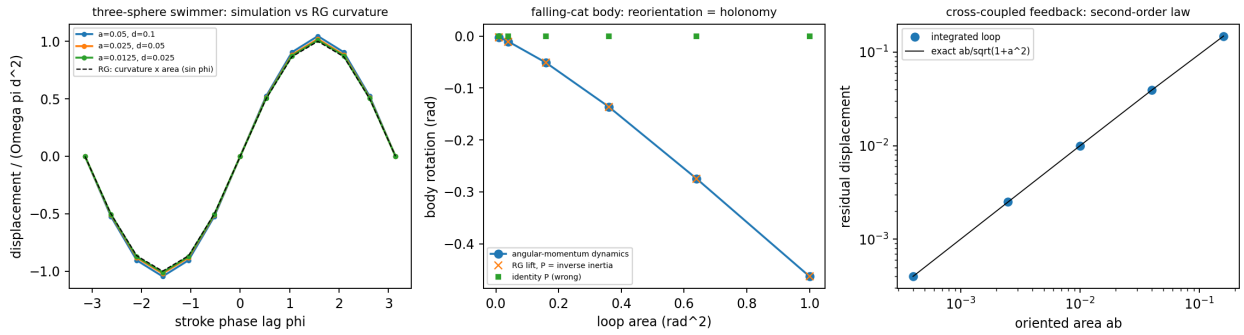


Figure 1: Left: the three-sphere swimmer's displacement, normalized by curvature times area, converges to $\sin\varphi$. Centre: falling-cat body rotation against loop area, comparing angular-momentum dynamics, the RG lift with $P = M^{-1}$, and the wrong (identity) cost. Right: the exact apparatus residual against oriented area.

8 Frozen holdout

1. *Calibrate.* Generate 120,000 local Gaussian increments at each of five shapes, and fit the bead radius and mobility scale within the declared Oseen family. No active outcome enters.

2. *Freeze.* Hash the calibration and the source files. A separate process writes predictions for 24 strokes: displacement, unit-power duration, loaded displacement and stall force.

3. *Score.* A later process runs the force-balance ODEs against declared gates. The displacement gate is $2 \times 10^{-7} + 0.08|\Delta X|$, and the duration gate is a relative error of 4%.

check	result
frozen unloaded displacement	24/24; max relative error 0.465%
frozen unit-power duration	24/24; max relative error 0.066%
frozen loaded displacement and stall	24/24; max stall residual 2.32×10^{-6}
power/energy balance	40 cases; max error 3.1×10^{-15}
power-controlled clock ODE	error 3.5×10^{-11}
Cartesian momentum	30 states; max error 3.9×10^{-11}
ODE refinement	max change 8.8×10^{-15}
canaries and negative controls	8/8 caught, plus the original 2/2
rerun from scratch	byte-identical frozen predictions

The canaries are:

- sign reversal, removed interaction and doubled response, each of which rejects all 24 predictions;
- post-freeze parameter or prediction edits, which break the hashes;
- negative controls for clock-scale omission, zero-area cancellation and omitted momentum drift.

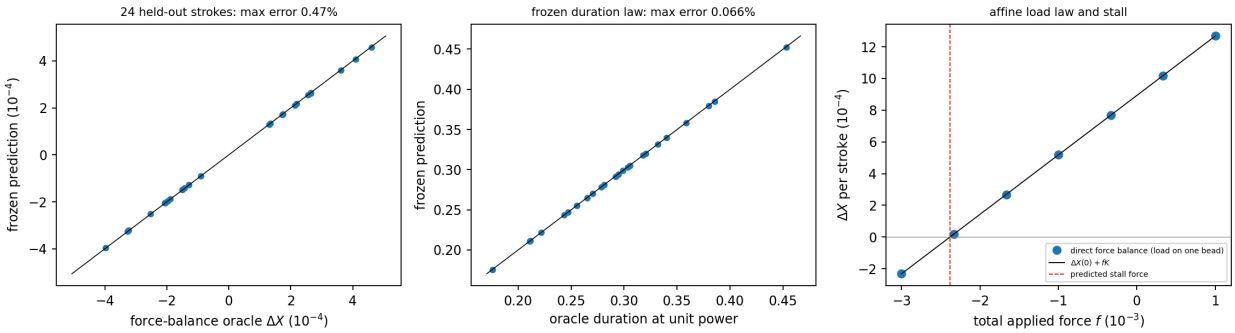


Figure 2: Left: frozen displacement predictions on 24 held-out strokes against the force-balance oracle. Centre: the frozen duration law against the oracle. Right: the affine load law from an independent force-balance swimmer loaded on one bead, with the predicted stall force.

Limits.

- This is a computational freeze-and-score holdout. It is not external blinding or tamper-proof preregistration.
- Predictor and oracle share the declared Oseen model. They use different equations, so their agreement checks a conditional correspondence and the implementation, not Oseen hydrodynamics in Nature.
- The calibration uses local covariance samples, not full Brownian trajectories. Finite-time drift, state-dependent stochastic calculus, measurement noise and misspecification are not validated.

9 What is RG's, and what is not

RG's.

- One object, $B = PJ^\top(JPJ^\top)^{-1}$, is simultaneously RG's canonical latent connection (L80), the noiseless Kalman lift, the minimum-dissipation motion of Stokes bodies, and the mechanical connection of free bodies.

- The Einstein relation makes the estimation and physics instances one, so fluctuations predict locomotion.
- The typing:
  - LOCKED (conditional) given $(F, P, \mathsf{H\text{-}resp})$;
  - CORRESPONDENCE without P (L79);
  - exact cancellation on reciprocal paths;
  - no clock from the connection alone;
  - no automatic descent of the dissipation metric.
- The protocol: calibrate locally, test H-resp locally, register, then run the loop.

Not RG's. The swimmer and falling-cat physics and their gauge description are established [9, 10, 15, 16], and overlap with geometric-phase theory is expected [17]. No priority is claimed.

Non-claims.

- Not spacetime curvature.
- The holonomy law is leading order in amplitude; it is not a force law.
- The physical inputs (cost, response rule and anchors) come from classical mechanics, hydrodynamics and statistical mechanics, not from GR or QM, and they are locally testable.
- All evidence is synthetic, within declared models.

10 Experimental programme

1. *The cross-coupled feedback apparatus:* the local H-resp test, then ΔZ.
2. *Colloidal three-bead swimmers* [13]:
   - calibrate from passive fluctuations, with an anchored prefactor;
   - predict displacement, duration at a fixed power and the stall force under an optical load;
   - check rate independence.
3. *Free-floating robots and satellites:* reorientation, and its drift when $L \neq 0$.

Each failure locates one of four things: H-resp, P, an anchor, or the declared model.

11 Status

statement	status
canonical lift, curvature, leading-order holonomy	theorem, given (J, P)
physical realization (a) and (b)	theorem, given the force or momentum balance
duration and energy law	theorem, given an absolute mobility and a power anchor
affine load law, stall force, momentum drift	theorem, given the declared balances
the laws on the declared synthetic models	supported (original battery, frozen holdout, audit)
absolute time; quotient-level metric	conditional on independent anchors or basicness
experimental adequacy	open

A Errata to the first edition

1. The first edition said that reciprocal strokes give zero "as flatness". The correct statement is *exact cancellation on reciprocal paths*. Zero signed area does not imply zero holonomy.

2. Reversal was stated to flip the sign in general. It gives the inverse holonomy; the sign flip is exact only at leading order, or in Abelian, symmetry-equivariant realizations (as for the planar body).

3. The exact-apparatus correction was called "cubic". It is cubic in A at fixed B, but quartic for $A = B$.

4. "1–1.5% on five seeds" was tied to the original runs only. Ten runs span 0.15–6.37%.

5. PC11 checks a status declaration; it is metadata, not an independent physical test.

6. "Temperature drops out" holds for the connection, not for absolute timing.

B Reproducibility

- The original `rg_physical_curvature_attack.py` is kept unchanged.
- `rg_physical_curvature_compatible.py` is the NumPy-2 compatible version, with the corrected wording. It passes 11/11, and its canaries are caught.
- `rg_lock_battery.py` (calibrate, predict, score) passes 18/18. Its `score` mode rescores the frozen artifacts without regenerating them.
- `rg_pcl_audit.py` (A01–A10) is the independent audit and passes 10/10.
- `rg_pcl_figure.py` draws the holdout figure from the saved records.

A rerun from scratch regenerates byte-identical frozen predictions.

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